



Department of Computer Science and Engineering

B.Tech. Computer Science and Engineering

Course Name : Human Computer Interaction
Course Code : TCS756
Semester : VII
Pre-requisite : Software Engineering (TCS611)

L	T	P	Credits
3	0	0	3

Course Learning Objectives:

- To introduce students to the fundamental concepts of Human Computer Interaction (HCI), human information processing, user interface design, and the importance of usability in interactive systems.
- To develop an understanding of HCI models, interaction styles, historical paradigms, and user interface concepts for designing effective and user-centered applications.
- To enable students to apply user-centered design processes, universal design principles, HCI standards, and interface design guidelines for developing intuitive and accessible user interfaces.
- To build analytical skills for evaluating user models, stakeholder requirements, socio-organizational issues, and user support mechanisms to improve the usability and effectiveness of interactive systems.
- To develop the ability to design and evaluate Human Computer Interaction systems through task analysis, dialog design, screen design, navigation techniques, and appropriate interaction methods to enhance user experience.

Course Outcomes:

After completion of the course, the students will be able to:

Course Outcome	Statement	Bloom's Level	Skill
CO1	Explain the capabilities of humans and computers from the viewpoint of human information processing and Human Computer Interaction fundamentals.	L2 – Understand	Understanding, Analytical Thinking
CO2	Describe Human Computer Interaction (HCI) models, interaction styles, historical paradigms, and user interface concepts.	L2 – understand	Understanding, Conceptual Thinking
CO3	Apply interactive design processes, universal design principles, HCI design principles, standards, and guidelines to develop effective user interfaces.	L3 – Apply	Design, Programming, Problem Solving
CO4	Analyse user models, user support, socio-organizational issues, and stakeholder requirements for designing effective HCI systems.	L4 – Analyze	Critical Thinking, Analytical Thinking, Problem Solving



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CO5	Design and evaluate HCI systems through task analysis, dialog design, screen design, navigation, and interaction techniques.	L6 – Create	Design, Creativity, Problem Solving, Evaluation
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Details of the Course:

Unit 1: Introduction

Contact Hours/Teaching Hours: 09

Introduction to Human Computer Interaction (HCI), Importance of User Interface, Definition and Importance of Good User Interface Design, Benefits of Good Design, A Brief History of Screen Design, Graphical User Interface (GUI), Popularity of Graphics, Concept of Direct Manipulation, Graphical Systems and their Characteristics, Web User Interface, Popularity and Characteristics of Web Interfaces, Principles of User Interface Design.

Self-Learning Hours = 6

Students shall explore the evolution of user interfaces, study examples of graphical and web-based interfaces, identify the principles of good user interface design, and evaluate the usability of commonly used software applications and websites.

Team Activity Hours = 3

Group discussion and presentation on comparing graphical user interfaces of different desktop and web applications, identifying good and poor design practices based on established user interface principles, and suggesting improvements to enhance usability.

Unit 2: Design Process

Contact Hours/Teaching Hours: 08

Design Process, Human Interaction with Computers, Importance of Human Characteristics, Human Considerations in Interface Design, Human Interaction Speeds, Understanding Business Functions.

Self-Learning Hours = 6

Students shall study human cognitive and physical characteristics that influence interface design, explore the stages of the user centred design process, analyse real-world user interfaces from usability and accessibility perspectives, and investigate the impact of human interaction speeds on system performance.

Team Activity Hours = 3

Students shall work in groups to analyse the design of existing software or web applications, identify human factors affecting usability, discuss business requirements influencing interface design, and present recommendations for improving the overall user experience.

Unit 3: Screen Designing

Contact Hours/Teaching Hours: 10

Design Goals, Screen Planning and Purpose, Organizing Screen Elements, Ordering of Screen Data and Content, Screen Navigation and Flow, Visually Pleasing Composition, Amount of Information, Focus and Emphasis, Presenting Information Simply and Meaningfully, Information Retrieval on the Web, Statistical Graphics, Technological Considerations in Interface Design.



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Self-Learning Hours = 6

Students shall explore the principles of effective screen design, evaluate the layout and navigation of existing desktop and web interfaces, study techniques for presenting information clearly through visual elements and statistical graphics, and examine technological factors influencing interface design.

Team Activity Hours = 3

Students shall work in groups to evaluate the usability of selected software or web interfaces, identify strengths and weaknesses in screen design, navigation, and information presentation, and propose improved interface layouts based on HCI design principles.

Unit 4: Windows

Contact Hours/Teaching Hours: 09

Windows – New and Navigation schemes selection of window, selection of devices based and screen-based controls. Components – text and messages, Icons and increases – Multimedia, colours, uses problems, choosing colours.

Self-Learning Hours = 6

Students shall study different types of windows, navigation schemes, and interface controls, explore the role of text, icons, multimedia, and colour in enhancing usability, and analyse the effectiveness of interface components used in popular desktop and web applications.

Team Activity Hours = 3

Students shall work in groups to evaluate the interface components of existing software applications, compare window navigation techniques and control mechanisms, assess the use of colours and multimedia elements, and recommend improvements based on usability and accessibility principles.

Unit 5: Software Tools

Contact Hours/Teaching Hours: 09

Software tools – Specification methods, interface – Building Tools. Interaction Devices – Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays. Drivers

Self-Learning Hours = 6

Students shall explore various interface specification and interface-building tools, study the working principles and applications of different interaction devices, investigate advancements in speech recognition and multimedia technologies, and analyse the role of device drivers in enabling effective human-computer interaction.

Team Activity Hours = 3

Students shall work in groups to compare different interface development tools and interaction devices, demonstrate the functionality of input and output devices used in HCI systems, and present their findings on emerging interaction technologies and their applications in modern user interfaces.



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Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO01	3	2								1		2	2	2	
CO02	2	2	1							2		2	2	2	1
CO03	2	2	3	2	3				1	2		2	3	2	2
CO04	1	3	2	2	2	2	1	1	2	2		2	3	2	2
CO05	2	3	3	3	3	1		1	2	2	1	2	3	2	3
Avg.	2	3	2	1	2	1	1	1	1	2	1	2	3	2	2

Text Books:

1. Wilbert O Galitz. (2016). *The essential guide to user interface design*”, 2nd Edition, Wiley Dreama Tech.
2. Ben Shneidermann (2009), *“Designing the user interface”*, 3rd Edition, Pearson Education Asia.

Reference Books:

1. Alan Dix, Janet Fincay, Gre Goryd, Abowd, Russell Bealg, (2003), *Human Computer Interaction*, Pearson Education, New Delhi, 3rd Edition.

Online Resources:

1. ILearn Platform GEHU- Human Computer Interaction (HCI)
<https://ilearn.gehu.ac.in/s/mycourses>
2. GeeksforGeeks – Human Computer Interaction (HCI)
<https://www.geeksforgeeks.org/human-computer-interaction-hci/>
3. NPTEL – Human Computer Interaction
<https://nptel.ac.in/>
4. Interaction Design Foundation (IDF) – Human-Computer Interaction
<https://www.interaction-design.org/>
5. Nielsen Norman Group (NN/g) – UX and Usability Resources
<https://www.nngroup.com/>
6. W3C Web Accessibility Initiative (WAI)
<https://www.w3.org/WAI/>